



Abhay Agarwal

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Bachelor of Technology

Electrical Engineering

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LeetCode | Codeforces

Education

Motilal Nehru National Institute of Technology Allahabad, Prayagraj

2024 – 2028

Bachelor of Technology in Electrical Engineering

CPI: 7.38/10.0

Experience

Software Engineer Intern (Backend + AI Systems Focused)

Mar 2026 – June 2026

EmployLabs.ai

Remote

- Engineered a **stateless HR AI Copilot** backend using Python and Anthropic API, featuring an intent-parsing orchestrator, **10 async agentic tools**, and conversation-history-driven state machines with regex fallback.
- Reduced HR candidate review time by **60%**, enabling teams to manage **3 times more job pipelines** simultaneously through **natural language actions** replacing manual dashboard navigation across more than 10 workflows.
- Built Zia's **Ring 2 memory** pipeline using PostgreSQL and Qdrant **semantic search**, 5-stage relationship tracker, door-rule memory enforcement engine, and Hinglish language calibrator for per-candidate AI personalization.
- Achieved sub-300ms context assembly and sub-2s end-to-end response latency across 30-35 turn conversations, with zero door-rule privacy violations in production.

Projects

AI-Driven Multi-Agent Platform for Campaign Planning and Optimization | [GitHub](#) Apr 2026 - May 2026

- Engineered **autonomous multi-agent system** using LangGraph-orchestrated directed graph coordinating **8 specialized LangChain agents** for end-to-end email campaign automation with human-in-the-loop approval gates and closed-loop feedback.
- Implemented **priority-queue optimization engine** scoring A/B variants via $\text{Score} = 0.7 \times \text{click-rate} + 0.3 \times \text{open-rate}$, auto-replacing the bottom 25% of performers with AI-regenerated content, validated against a **10,000-row synthetic customer dataset**.
- Tech Stack:** Python, deepseek-v4-pro, LangChain, LangGraph, FastAPI, MongoDB

Context-Aware Academic Assistant Using Hybrid RAG Architecture | [GitHub](#)

Dec 2025 - Feb 2026

- Designed **hybrid RAG system** combining Qdrant vector search and Neo4j graph traversal with Cohere reranking across **50K+ document chunks**, achieving **89% top-3 accuracy** on 70+ exam questions.
- Applied **Self-RAG critique agents** to cut hallucination rate from 31% to 8% on a held-out eval set, maintaining sub-200ms P95 retrieval latency and **0.91 faithfulness on RAGAS**.
- Tuned RAG pipeline hyperparameters (chunk size, top-k, reranking threshold) via grid search, improving top-3 retrieval accuracy
- Tech Stack:** Python, FastAPI, LangChain, OpenAI GPT-5.1, Qdrant, NeonDB, Neo4j, RAGAS, LangSmith

Vision-Based Lane Understanding and Intelligent Driving Decision Engine | [GitHub](#)

Jan 2026 - Mar 2026

- Constructed a vision-based autonomous driving decision engine for unlabeled campus road images using a **hybrid geometric + ML pipeline** with pseudo-labeling, achieving **88% command accuracy** (Forward/Left/Right) and **99% STOP precision** and preprocessing the training images with a Four-Gate Image Quality Filter.
- Implemented **defense-in-depth safety using 5-Layered architecture** with MC Dropout uncertainty quantification, temporal consistency filtering, and **GradCAM explainability** overlays for every prediction
- Tech Stack:** Python, PyTorch, MiDaS, UltraFast Lane Detection v2, YOLOv8, FastAPI, ONNX, MobileNetV3

Skills

Languages: Java, Python, SQL **Database Systems:** MongoDB, PostgreSQL, MySQL, Neo4j, Qdrant

Frameworks and Tools: scikit-learn, PyTorch, LangGraph, LangChain, LangSmith, RAGAS, FastAPI, Docker

CS core: OOPS, Data Structures and Algorithms, Operating System, DataBase Management System

Achievements

Competitive Programming: Rated as Specialist on Codeforces with a peak rating of 1412

First Runner-up: HACKATRON, technical fest (Infotsav) held at IIITM Gwalior

Winner: Logical Rhythms (Codesangam 2025), Machine Learning event by CC Club, MNNIT

Special Mention: Dev or Die (Power Surge 2025), for developing the **Command Nest**, a multi-doc retrieval system

Winner: Robowars (BotRush 2025), 1v1 robot fight event by Robotics Club, MNNIT

Winner: DoodleBot (Robomania 2024), path tracing via bot event by Robotic Club MNNIT